STABLECOINS AND THE EXORBITANT PRIVILEGE

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Topics in International Finance
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Roadmap

This week:

- ❖ Motivation — why stablecoins matter for sovereign debt
- ❖ The New Triffin Dilemma — our core argument
- ❖ Theoretical extension of Maggiori (2017)
- ❖ Data & Methodology
- ❖ Result 1 — Privilege amplification regression
- ❖ Result 2 — Reserve adequacy threshold (Hansen 2000)

Next week:

- ❖ Result 3 — Buffer-conditioned event study
- ❖ Conclusion & Policy implications

Motivation: Stablecoins as Treasury Buyers

$300bn+

Total USD-pegged supply (2026)

$127bn

Tether T-bill holdings (Q2 2025)

$1–2 tn

Projected ecosystem size

4

Jurisdictions finalising legislation

The same structural role that deepens US safe-asset demand also embeds a fragility that can rapidly reverse it.

Exorbitant Privilege and the Original Triffin Dilemma

Exorbitant Privilege

- *The US issues the world's reserve currency — allowing it to borrow cheaply while investing in higher-yielding foreign assets*
- *This "privilege" is measured through OIS–Treasury spread compression: when demand for T-bills rises, US borrowing costs fall below the market baseline*
- *USD-pegged stablecoins now replicate this channel privately — creating $300bn+ in structural Treasury demand outside sovereign control*

Research Topic

Stablecoins and Exorbitant Privilege: A New Channel for Safe-Asset Demant and Its Systematic Fragility

Triffin Dilemma (1960)

A reserve currency country must supply sufficient safe assets to meet global demand — but this requires running structural current account deficits that progressively undermine confidence in the currency itself. The system depends on the very condition that makes it unsustainable.

The New Triffin Dilemma

Our analogue for stablecoin issuers:

Normal times ✅

- Stablecoin issuance → mechanical T-bill acquisition by issuers
- Structural increase in safe-asset demand for US Treasuries
- OIS–Treasury spread compression: yield falls below SOFR baseline
- Exorbitant privilege amplified via private intermediation channel

Run scenario ❌

- Redemption shock → programmatic T-bill liquidation
- Treasury yields spike above OIS — spread reversal
- Sovereign-linked liquidity shock transmitted to debt markets
- Self-fulfilling dynamic: fear validates liquidation, crisis propagates

What determines outcome

- θ = Treasury holdings / Supply → structural demand channel
- L = Liquid reserves / Supply → crisis absorption capacity
- $L \geq L^*$ → redemptions absorbed internally, no market spillover
- $L < L^*$ → forced liquidation, sovereign spread disruption
- We estimate the threshold L^* (Hansen 2000)

In our paper: We separate θ (why stablecoins help the US) from L (what prevents a crisis)

Theoretical Extension: Maggiori (2017)

Maggiori (2017) original:

Safe-asset demand driven by Ω^* — the countercyclical marginal value of RoW financial net worth.

Our extension adds S, θ, and L:

$$D^*(\tilde{N}^*, S, \vartheta, L) = D^*precautionary(\tilde{N}^*) + D^*stablecoin(S, \vartheta) + D^*buffer(L)$$

Safe-asset demand now has three components:

- D*precautionary — Maggiori's original term, driven by global wealth
- D*stablecoin — new structural demand via mechanical T-bill acquisition
- D*buffer — additional demand from excess reserve holdings above minimum

Main Estimating Equation

$$\begin{aligned} Spread_t = \alpha &+ \beta_1 \cdot \Delta \ln S_t + \beta_2 \cdot \theta_t + \beta_3 \cdot L_t \\ &+ \beta_4 \cdot (L_t \times \Delta \ln S_t) \\ &+ \beta_5 \cdot V_t + \beta_6 \cdot VIX_t \\ &+ \beta_7 \cdot \Delta \ln N^*_t + \varepsilon_t \end{aligned}$$

Predicted signs:

- $\beta_1 < 0$: ΔlnS compresses spreads (privilege amplification)
- $\beta_2 > 0$: Higher T-bill exposure θ amplifies demand
- $\beta_4 > 0$: Higher L dampens crisis transmission during runs

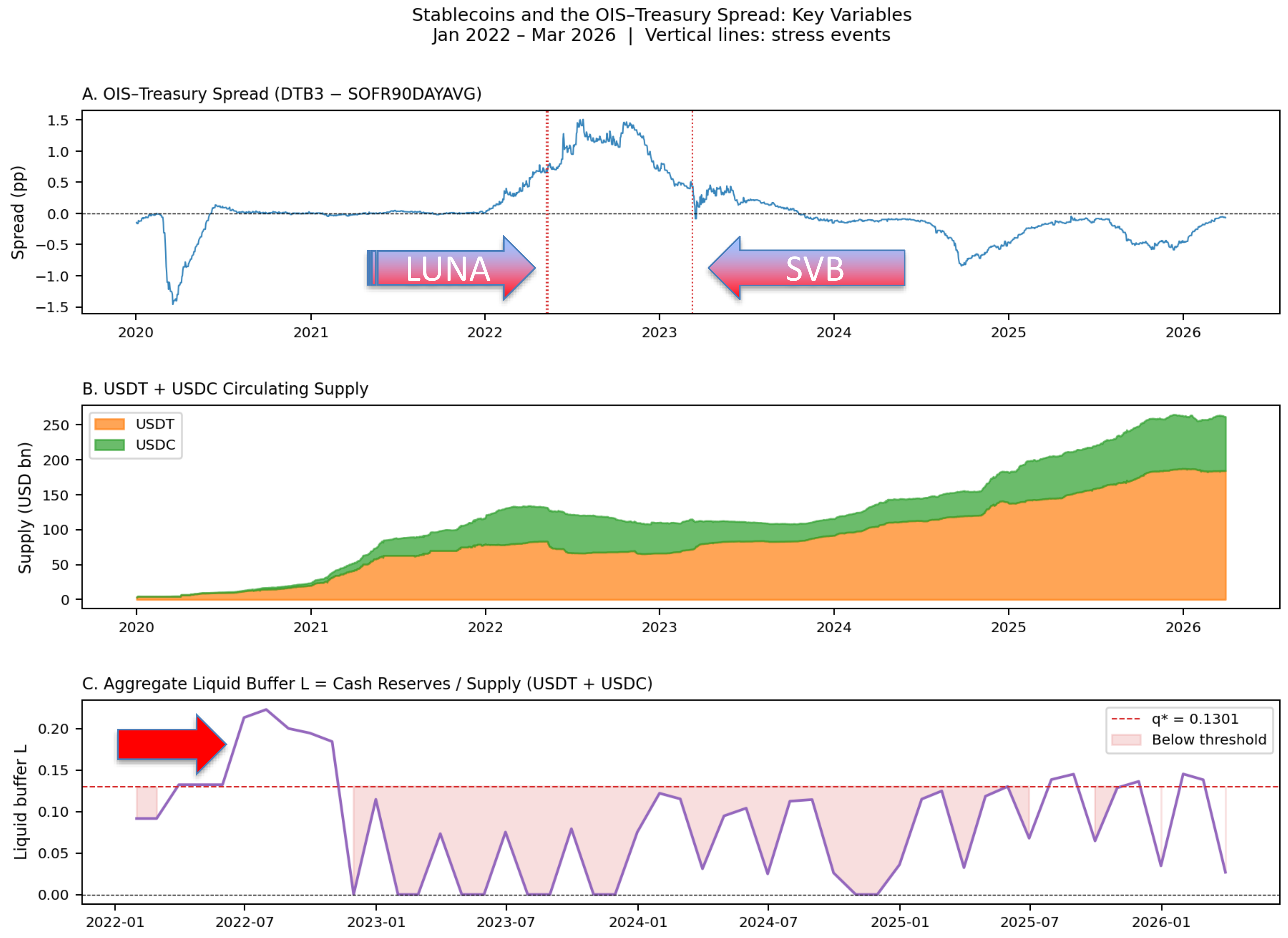
The buffer L introduces a non-linear threshold — the key theoretical departure from Maggiori's original model.

Data & Sample

Variable	Definition	Source	Frequency
OIS–Treasury Spread	DTB3 – SOFR90DAYAVG	FRED (direct CSV, no API key)	Daily → Monthly
Stablecoin Supply S	USDT + USDC market cap	DeFiLlama stablecoins API (free)	Daily → Monthly
θ (Treasury Exposure)	T-bill holdings / supply	Tether/BDO + Circle/Deloitte attestations	Quarterly/Monthly
L (Liquid Buffer)	Cash reserves / supply	Tether/BDO + Circle/Deloitte attestations	Quarterly/Monthly
Velocity V	7-day rolling SD of $\Delta \ln S$	Computed from DeFiLlama	Daily → Monthly
VIX	CBOE VIX index	FRED	Daily → Monthly
$\Delta \ln N^*$ (RoW equity)	ACWX ETF log-return	Yahoo Finance	Daily → Monthly

Sample: January 2022 – March 2026 | N = 51 monthly observations | ϑ and L sourced from Tether/BDO and Circle/Deloitte reserve attestations

Key Variables: January 2022 – March 2026



Three panels

❖ A: Treasury spread — compresses when supply grows, spikes in crises.

T-bill (DTB3) is a **3-month U.S. government bond**. **Secured Overnight Financing Rate** is the rate banks charge for one night.

❖ Red vertical lines = stress events we study

❖ B: USDT + USDC supply — grew from $100bn to $200bn+ over this period

❖ C: Liquid buffer L — dangerously thin in 2022, recovering since 2023

❖ Red dashed line = our estimated safety threshold (13%)

What the Data Shows Us

MORE SUPPLY → LOWER SPREAD

As USDT and USDC supply grew, the gap between Treasury yields and overnight rates compressed. Stablecoin issuers were buying U.S. T-bills to back their tokens — deepening demand for safe assets.

CRISIS → SPREAD SPIKES

During the LUNA/UST collapse (May 2022), spreads spiked as issuers were forced to liquidate T-bill holdings to meet redemptions. The privilege reversed — exactly what our theory predicts.

BUFFER WAS THIN WHEN IT MATTERED MOST

In 2022, liquid cash reserves sat below our estimated 13% safety threshold. Issuers had to sell T-bills instead of using cash — amplifying the shock.

Result 1: Stablecoin Growth Compresses Treasury Spreads

H1 CONFIRMED — statistically significant at the 5% level

$$\beta_1 = -6.02^{**}$$

N = 51 months | $R^2 = 0.50$

What this means in plain terms:

Every 1% growth in stablecoin supply compresses the Treasury spread by about **6 basis points.**

When Tether and Circle expand, they buy more U.S. T-bills — pushing Treasury yields down relative to overnight rates. This is the exorbitant privilege being amplified.

Variable	Coefficient	Sig.
$\Delta \ln S$ (stablecoin supply growth)	**−6.017**	**
θ (Treasury Exposure)	0.100	—
L (Liquid Buffer)	1.034	—
$L \times \Delta \ln S$ (buffer interaction)	3.891	—
Controls (velocity, VIX, $\Delta \ln N^*$)	included	

Not significant enough

*** $p < 0.05$ — = not significant at conventional levels HAC Newey–West SE (3 lags)*

Result 1: When Data Quality Improves, the Picture Sharpens

The buffer variables (ϑ and L) were not individually significant at our full sample of 51 months. This is a power problem — not a theory problem. Reserve attestations only became comprehensive after 2023.

When we restrict to January 2023 – March 2026 (N = 39 months, best attestation coverage):

Supply channel

$$\beta_1 = -8.14$$

***

Even stronger supply effect in recent data. Confirmed at the 1% level.

Buffer interaction

$$L \times \Delta \ln S = +49.17$$ *** (p = 0.004)

$\beta_1 + \beta_4 \times L = -8.14 + 49.17 \times L$

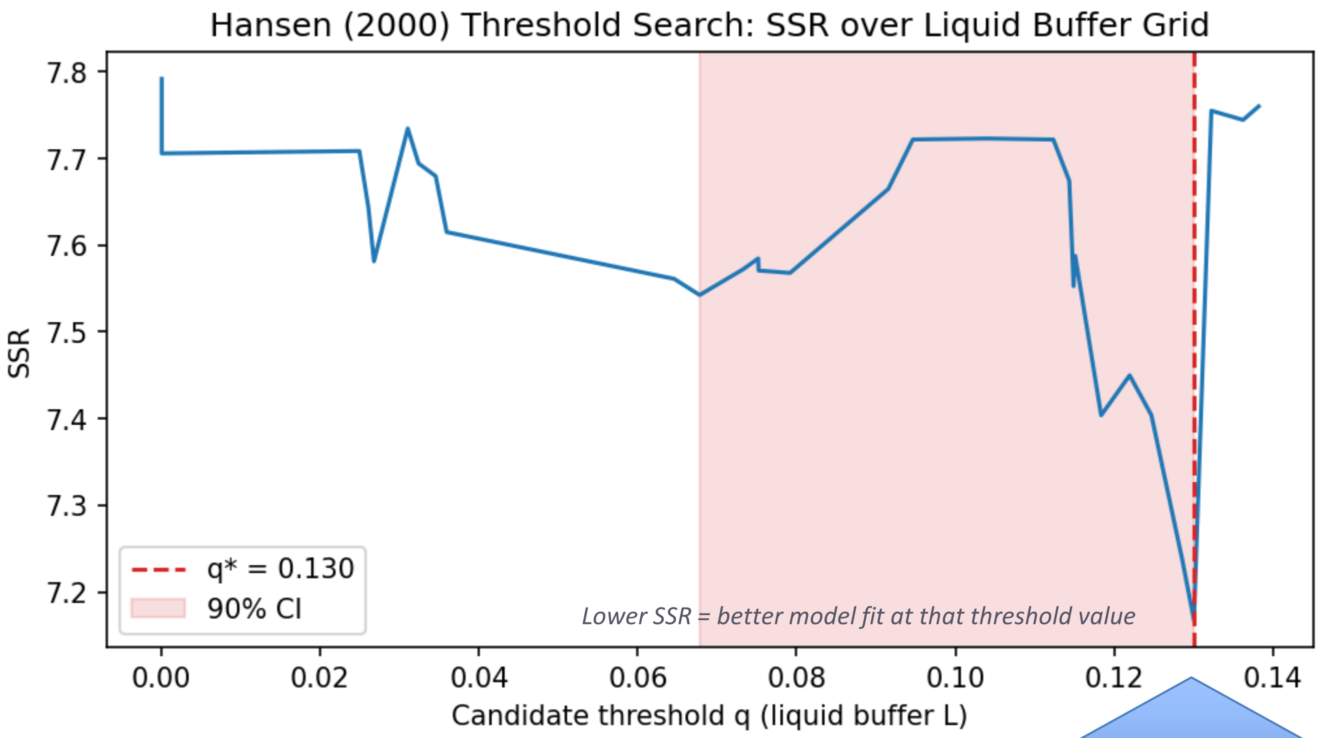
When L is **low** (say 0.05): total effect = $-8.14 + (49.17 \times 0.05) =$ **-5.68** → strong compression, issuers are actively buying T-bills

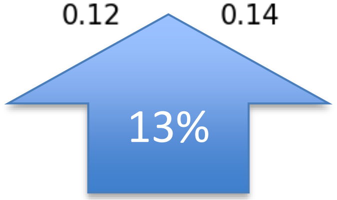
When L is **high** (say 0.16): total effect = $-8.14 + (49.17 \times 0.16) =$ **-0.27** → effect nearly disappears, issuers have enough cash and don't need to move T-bill markets

Both channels confirmed in recent data. The mechanism is real — our full sample just needs more months to reach significance.

Result 2: Is There a Safety Threshold?

SSR : Sum of Squared Residuals





$q^* = 13\%$

of supply held as liquid cash

What this means:

Using Hansen (2000) threshold regression, we find a tipping point at 13% liquid cash reserves.

Below this level — where most of our data falls — issuers don't have enough cash to absorb redemptions without selling T-bills.

Bootstrap p = 0.260 (suggestive — the direction is clear; significance grows as data accumulates past 2026)

Below vs. Above the Threshold: Two Different Worlds

Splitting our 51 months at q = 13% reveals a striking difference in how stablecoin growth affects Treasury markets:*

BELOW 13% LIQUID BUFFER

38 out of 51 months

β = −6.97

"With no cash buffer, every new token issued requires buying T-bills. That constant T-bill demand compresses spreads strongly. And during redemptions, they have no cash — so they must sell T-bills too. ***Their activity moves the market in both directions."***

ABOVE 13% LIQUID BUFFER

13 out of 51 months

β = +1.26

"With enough cash on hand, new tokens are backed by existing reserves — no T-bill buying needed. During redemptions, cash absorbs the shock instead of T-bill selling. ***The T-bill market is undisturbed."***

The liquid cash buffer is what separates a stablecoin issuer that stabilises Treasury markets from one that disrupts them.

Looking Ahead: Causal Proof & Policy Solutions

- **Result 3: The Buffer-Conditioned Event Study**
    - Moving from monthly trends to daily "quasi-experimental" causal identification.
    - Analyzing three major stress episodes: LUNA/UST, the USDT partial depeg, and the USDC/SVB failure.
    - Proving how the **buffer size** determines if a crisis causes a "sovereign-linked liquidity shock" or a "flight-to-safety".

- **Robustness: "Data Sharpening" Post-2023**
    - Testing a more recent sub-sample (N=39) where reporting and attestations are more transparent.
    - Showing how the relationship between reserves and market spreads becomes even more statistically significant as the market matures.

- **Policy Recommendation: The "13% Liquid Reserve Rule"**
    - Translating the threshold regression into a concrete regulatory requirement.
    - Proposing a **minimum 13% liquid buffer** (cash/near-cash) to prevent private stablecoin runs from spilling into public Treasury markets

- **Conclusion: Navigating the New Triffin Dilemma**
    - Final synthesis of how stablecoins both support and threaten the U.S. "exorbitant privilege".

THANK YOU

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